	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary Summer 2024 **Course: B. Tech.** **Branch: All** **Semester: II** **Subject Code & Name: BTBS202P Engineering Physics** **Max Marks: 60** **Date:14/06/2024** **Duration: 3 Hr.**		
	Instructions to the Students: *1. All the questions are compulsory.* *2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.* *3. Use of non-programmable scientific calculators is allowed.* *4. Assume suitable data wherever necessary and mention it clearly.*		
		(Level/CO)	**Marks**
Q. 1	**Solve Any Two of the following.**		**12**
A)	Define Damped oscillations. Derive an expression for differential equation of Damped oscillations.	(CO1) (Remember & Understand)	**6**
B)	Define i) Magnetostriction Effect ii) Piezoelectric Effect iii) Resonance. Calculate the natural frequency of the ultrasonic waves generated by a quartz crystal having thickness of 5.5 mm. [Given Y= 8×10^{10} N/m^2, ρ (density)=2650 Kg/m^3]	(CO1) (Remember & Understand)	**6**
C)	Derive differential equation of wave motion.	(CO1) Understand)	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Derive an expression for diameter of n th bright and dark Newton's ring.	(CO2) (Understand)	**6**
B)	Explain the construction & working of Helium Neon Laser with neat, labeled diagram.	(CO2) (Understand)	**6**
C)	State any six applications of optical fiber. Numerical aperture of a fiber is 0.5 and core refractive index is 1.48. Find cladding refractive index.	(CO2) (Remember & Understand)	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Explain the construction & working of Bainbridge mass spectrograph with neat & labeled diagram.	(CO3) (Understand)	**6**
B)	Explain the construction & working of Geiger Muller Counter.	(CO3) (Understand)	**6**

C)	Derive Schrodinger’s time independent wave equation.	(CO3) (Understand)	**6**
Q.4	**Solve the following questions.**		**12**
A)	Calculate atomic radii for Simple cubic, Body centered, and face centered cubic structure.	(CO4) (Understand)	**6**
B)	Explain Continuous X-ray spectra. Prove that, $$\lambda_{min} = \frac{12400}{V} A^0$$	(CO4) (Understand)	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Explain B-H Curve for ferromagnetic materials. Define the terms Retentivity & Coercivity.	(Understand)	**6**
B)	Explain Type-I & Type II superconductors.	(Understand)	**6**
C)	Explain Conductor, Semiconductor, and Insulator on the basis of band theory of Solids.	(Understand)	**6**
	***** End *****		